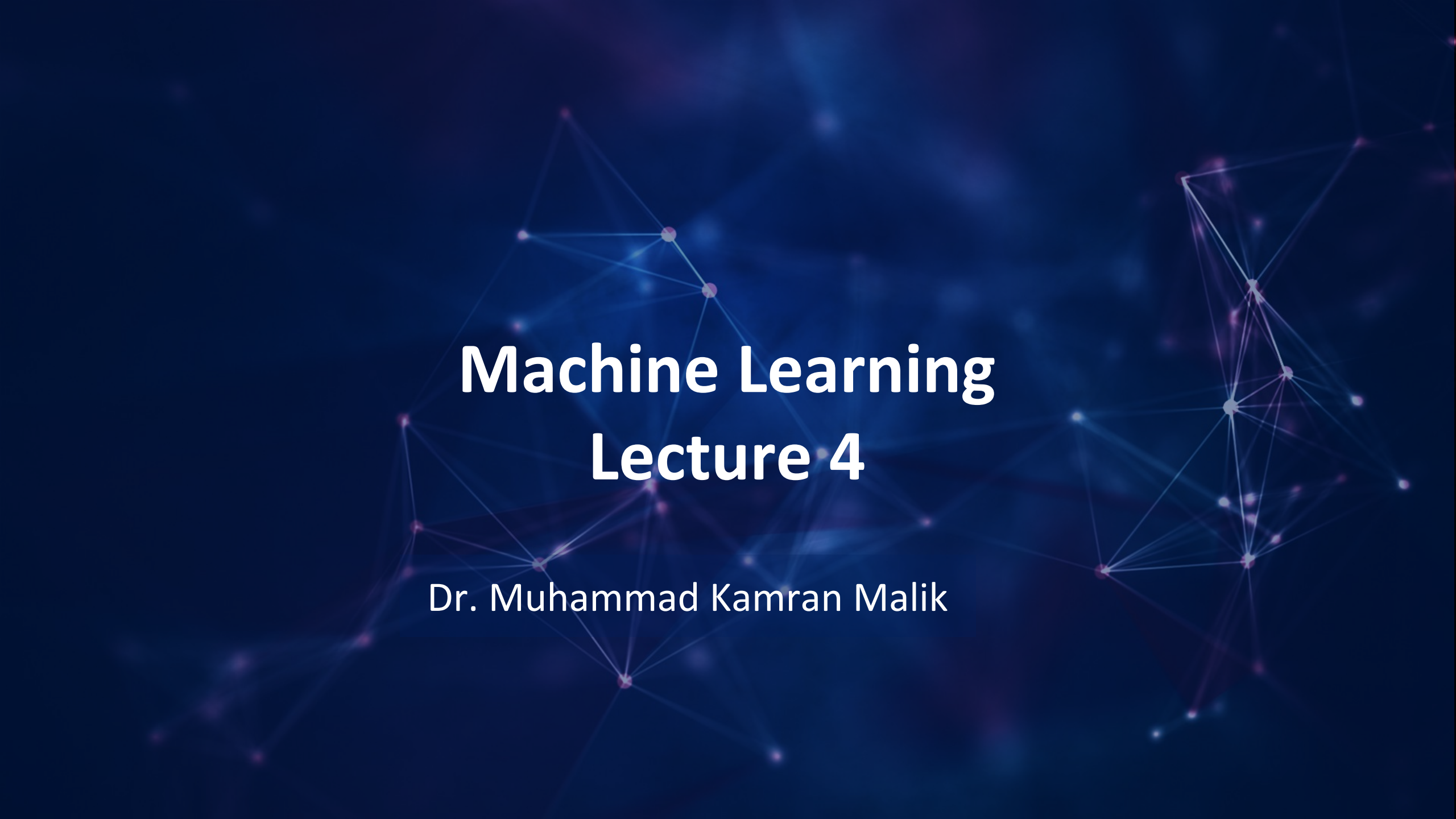




Machine Learning Lecture 4

Dr. Muhammad Kamran Malik



Linear Regression using Simultaneous equation

Linear Regression with One Variable

The Hypothesis Function / Model

- $y' = h_{\vartheta}(x) = \vartheta_0 + \vartheta_1 x$
- $y'^{(i)} = h_{\vartheta}(x^{(i)}) = \vartheta_0 + \vartheta_1 x^{(i)}$

Cost function:

$$J(\vartheta_0, \vartheta_1) = \frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2$$

Objective:

$$\min_{\vartheta_0, \vartheta_1} J(\vartheta_0, \vartheta_1)$$

$$\begin{aligned} \frac{\partial}{\partial \vartheta_0} J(\vartheta_0, \vartheta_1) &= \frac{\partial}{\partial \vartheta_0} \left(\frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2 \right) \\ &= \frac{\partial}{\partial \vartheta_0} \left(\frac{1}{2m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)})^2 \right) \\ &= \frac{2}{2m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) \frac{\partial}{\partial \vartheta_0} (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) \\ &= \frac{1}{m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) \\ \frac{1}{m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) &= 0 \\ \Rightarrow \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) &= 0 \\ \Rightarrow \sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} - \sum_{i=1}^m y^{(i)} &= 0 \\ \Rightarrow \sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} = \sum_{i=1}^m y^{(i)} &\text{-----> A} \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \vartheta_0} J(\vartheta_0, \vartheta_1) &= 0 \\ \frac{\partial}{\partial \vartheta_1} J(\vartheta_0, \vartheta_1) &= 0 \end{aligned}$$



Q & A

Linear Regression with One Variable

The Hypothesis Function / Model

- $y' = h_{\vartheta}(x) = \vartheta_0 + \vartheta_1 x$
- $y'^{(i)} = h_{\vartheta}(x^{(i)}) = \vartheta_0 + \vartheta_1 x^{(i)}$

Cost function:

$$J(\vartheta_0, \vartheta_1) = \frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2$$

Objective:

$$\min_{\vartheta_0, \vartheta_1} J(\vartheta_0, \vartheta_1)$$

$$\frac{\partial}{\partial \vartheta_0} J(\vartheta_0, \vartheta_1) = \frac{\partial}{\partial \vartheta_0} \left(\frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2 \right)$$

$$= \frac{\partial}{\partial \vartheta_0} \left(\frac{1}{2m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)})^2 \right)$$

$$= \frac{2}{2m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) \frac{\partial}{\partial \vartheta_0} (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)})$$

$$= \frac{1}{m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)})$$

$$\frac{1}{m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) = 0$$

$$\Rightarrow \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} - y^{(i)}) = 0$$

$$\Rightarrow \sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} - \sum_{i=1}^m y^{(i)} = 0$$

$$\Rightarrow \sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} = \sum_{i=1}^m y^{(i)} \quad \text{-----> A}$$

$$\sum_{i=1}^m \vartheta_0 x^{(i)} + \sum_{i=1}^m \vartheta_1 x^{2(i)} = \sum_{i=1}^m y^{(i)} x^{(i)} \quad \text{-----> B}$$

$$\begin{aligned} \frac{\partial}{\partial \vartheta_0} J(\vartheta_0, \vartheta_1) &= 0 \\ \frac{\partial}{\partial \vartheta_1} J(\vartheta_0, \vartheta_1) &= 0 \end{aligned}$$

Linear Regression using simultaneous equation

- $\sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} = \sum_{i=1}^m y^{(i)}$ -----> A
- $\sum_{i=1}^m \vartheta_0 x^{(i)} + \sum_{i=1}^m \vartheta_1 x^{2(i)} = \sum_{i=1}^m y^{(i)} x^{(i)}$ -----> B

$$8x + 2y = 46$$

$$7x + 3y = 47$$

A X B

$$\bullet \begin{bmatrix} \sum_{i=1}^m 1 & \sum_{i=1}^m x^{(i)} \\ \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} \end{bmatrix} \begin{bmatrix} \vartheta_0 \\ \vartheta_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^m y^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{(i)} \end{bmatrix}$$

$$\begin{bmatrix} 8 & 2 \\ 7 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 46 \\ 47 \end{bmatrix}$$

$$\bullet \begin{bmatrix} m & \sum_{i=1}^m x^{(i)} \\ \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} \end{bmatrix} \begin{bmatrix} \vartheta_0 \\ \vartheta_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^m y^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{(i)} \end{bmatrix}$$



Q & A



Training a linear regression model



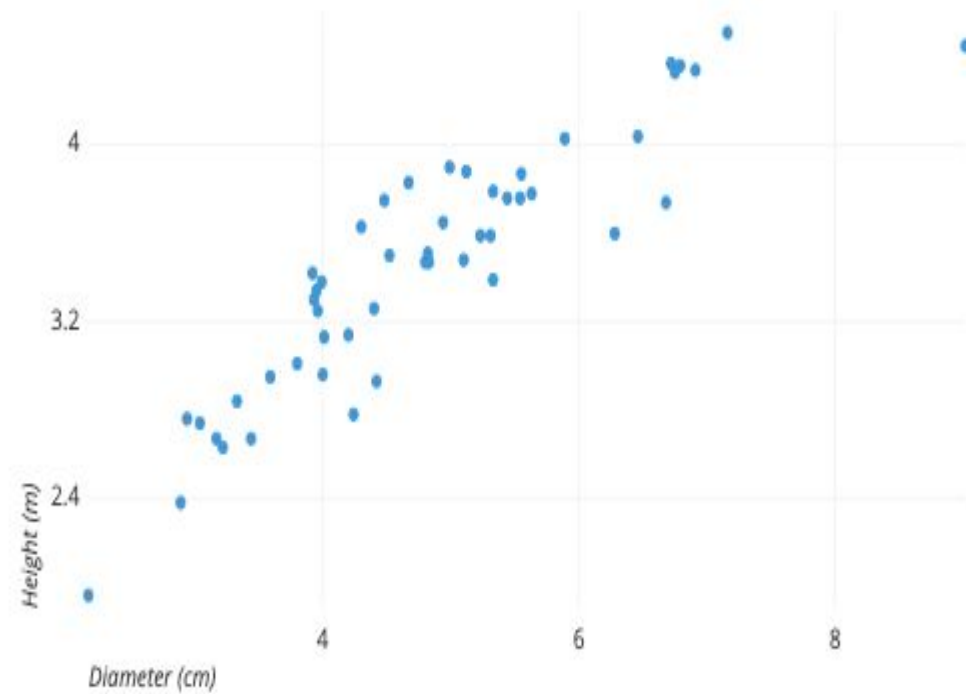
A

X

B

$$\begin{bmatrix} m & \sum_{i=1}^m x^{(i)} \\ \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} \end{bmatrix} \begin{bmatrix} \vartheta_0 \\ \vartheta_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^m y^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{(i)} \end{bmatrix}$$

ht (x)	wt (y)
1	10
2	20
3	30
4	40



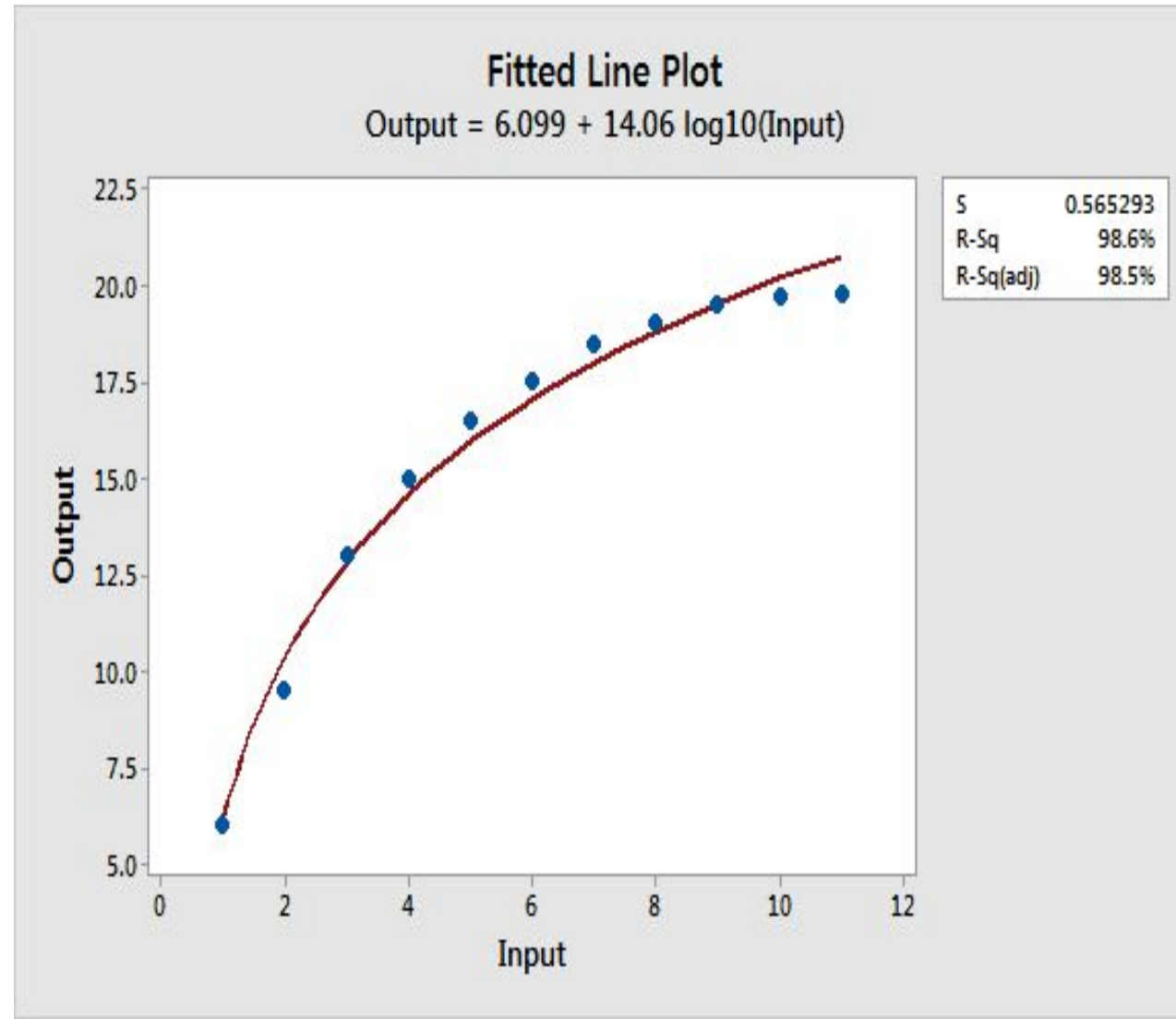


Q & A



Training a higher degree models

Quadratic Model



Quadratic Model

The Hypothesis Function / Model

- $y' = h_{\vartheta}(x) = \vartheta_0 + \vartheta_1 x + \vartheta_2 x^2$
- $y'^{(i)} = h_{\vartheta}(x^{(i)}) = \vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)}$

Cost function:

$$J(\vartheta_0, \vartheta_1, \vartheta_2) = \frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2$$

Objective:

$$\min_{\vartheta_0, \vartheta_1, \vartheta_2} J(\vartheta_0, \vartheta_1, \vartheta_2)$$

$$\frac{\partial}{\partial \vartheta_0} J(\vartheta_0, \vartheta_1, \vartheta_2) = \frac{\partial}{\partial \vartheta_0} \left(\frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2 \right)$$

$$= \frac{\partial}{\partial \vartheta_0} \left(\frac{1}{2m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)} - y^{(i)})^2 \right)$$

$$= \frac{2}{2m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)} - y^{(i)}) \frac{\partial}{\partial \vartheta_0} (\vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)} - y^{(i)})$$

$$= \frac{1}{m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)} - y^{(i)})$$

$$\frac{1}{m} \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)} - y^{(i)}) = 0$$

$$\Rightarrow \sum_{i=1}^m (\vartheta_0 + \vartheta_1 x^{(i)} + \vartheta_2 x^{2(i)} - y^{(i)}) = 0$$

$$\Rightarrow \sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} + \sum_{i=1}^m \vartheta_2 x^{2(i)} - \sum_{i=1}^m y^{(i)} = 0$$

$$\Rightarrow \sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} + \sum_{i=1}^m \vartheta_2 x^{2(i)} = \sum_{i=1}^m y^{(i)} \text{ -----> A}$$

$$\frac{\partial}{\partial \vartheta_0} J(\vartheta_0, \vartheta_1, \vartheta_2) = 0$$

$$\frac{\partial}{\partial \vartheta_1} J(\vartheta_0, \vartheta_1, \vartheta_2) = 0$$

$$\frac{\partial}{\partial \vartheta_2} J(\vartheta_0, \vartheta_1, \vartheta_2) = 0$$

Linear Regression using simultaneous equation

- $\sum_{i=1}^m \vartheta_0 + \sum_{i=1}^m \vartheta_1 x^{(i)} + \sum_{i=1}^m \vartheta_2 x^{2(i)} = \sum_{i=1}^m y^{(i)}$ -----> A
- $\sum_{i=1}^m \vartheta_0 x^{(i)} + \sum_{i=1}^m \vartheta_1 x^{2(i)} + \sum_{i=1}^m \vartheta_2 x^{3(i)} = \sum_{i=1}^m y^{(i)} x^{(i)}$ -----> B
- $\sum_{i=1}^m \vartheta_0 x^{2(i)} + \sum_{i=1}^m \vartheta_1 x^{3(i)} + \sum_{i=1}^m \vartheta_2 x^{4(i)} = \sum_{i=1}^m y^{(i)} x^{2(i)}$ -----> C

$$8x + 2y + 3z = 46$$

$$7x + 3y + 4z = 47$$

$$2x + y + 2z = 1$$

$$\begin{matrix} A & X & B \end{matrix}$$

$$\begin{bmatrix} 8 & 2 & 3 \\ 7 & 3 & 4 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 46 \\ 47 \\ 1 \end{bmatrix}$$

$$\begin{aligned} & \bullet \begin{bmatrix} \sum_{i=1}^m 1 & \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} \\ \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} & \sum_{i=1}^m x^{3(i)} \\ \sum_{i=1}^m x^{2(i)} & \sum_{i=1}^m x^{3(i)} & \sum_{i=1}^m x^{4(i)} \end{bmatrix} \begin{bmatrix} \vartheta_0 \\ \vartheta_1 \\ \vartheta_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^m y^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{2(i)} \end{bmatrix} \\ & \bullet \begin{bmatrix} m & \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} \\ \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} & \sum_{i=1}^m x^{3(i)} \\ \sum_{i=1}^m x^{2(i)} & \sum_{i=1}^m x^{3(i)} & \sum_{i=1}^m x^{4(i)} \end{bmatrix} \begin{bmatrix} \vartheta_0 \\ \vartheta_1 \\ \vartheta_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^m y^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{2(i)} \end{bmatrix} \end{aligned}$$



Q & A



Programming Assignment

Programming Assignment

- Import libraries

```
import pandas as pd
import numpy as np
%matplotlib inline
import matplotlib.pyplot as plt
```

- Load both csv files (RegressoinTesting.csv, and RegressoinTraining.csv)

```
Trn_Data = pd.read_csv('RegressoinTraining.csv')
```

- Show top five training and testing records

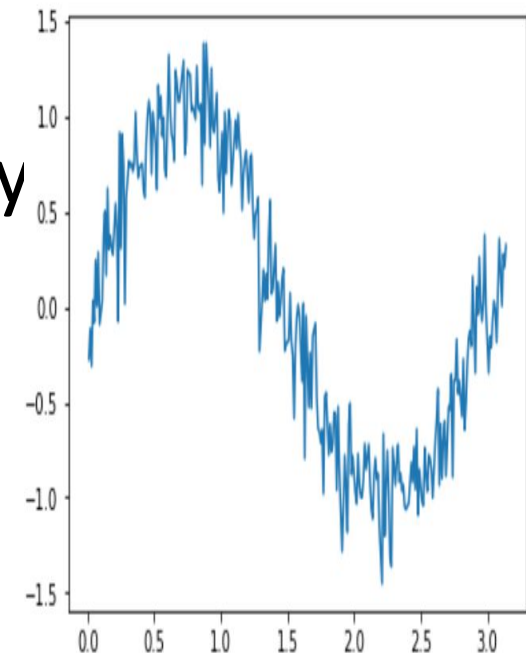
```
Trn_Data.head() # show top five records
```

- Type cast training data and testing data into numpy

```
x = np.array(Trn_Data['sales'])
```

```
y = np.array(Trn_Data['profit'])
```

- Plot training and testing data using (plt.plot (x, y))



Programming Assignment

- Fit linear regression models to the **training data only**

$$\begin{bmatrix} m & \sum_{i=1}^m x^{(i)} \\ \sum_{i=1}^m x^{(i)} & \sum_{i=1}^m x^{2(i)} \end{bmatrix} \begin{bmatrix} \vartheta_0 \\ \vartheta_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^m y^{(i)} \\ \sum_{i=1}^m y^{(i)} x^{(i)} \end{bmatrix}$$

Cost function:

$$J(\vartheta_0, \vartheta_1) = \frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2$$

Training

Data x	y
1	2.3
1.2	2.0
1.1	1.3
...	...

Testing

Data tst_x	tst_y
2	3.4
2.1	3.2
2.1	2.2
...	...



Q & A